

Marlon Buchanan

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Industry Experience

R&D Engineering Co-op, FTXT Energy Technology – North Vancouver, BC Sept 2023 – Apr 2024

- Characterized fuel cell plate resistance under compression using a UTM, recording voltage drop per layer for thermal resistance network modeling
- Automated life-cycle mechanical analysis in MATLAB, generating 120+ FEA load profiles for GB/T 28046.3 compliance, reducing analysis time by 50%
- Built a Python scraper to populate a materials database (75+ entries), reducing material selection research time by 20%
- Verified GD&T compliance on 5+ parts against technical drawings, catching a dimensional fault before shock and vibration testing

Engineering Projects

Self-Balancing Robot (Edmond) - STM32, C, MATLAB/Simulink, SolidWorks Aug 2026 – Present

- Designed and 3D-printed a SolidWorks chassis integrating motors, electronics, and OLED; robot balances unaided and rejects disturbances up to $\pm 10^\circ$
- Designed a discrete LQR controller in MATLAB/Simulink on a linearized cart-pendulum model, with physical parameters from SolidWorks mass properties
- Implemented complementary filter sensor fusion on STM32F303K8 in HAL C, fusing MPU6050 accel/gyro data over I²C for real-time pitch estimation
- Diagnosed intermittent I²C failures from loose connectors; wrote a modular driver with bus-recovery logic and soldered the final build to eliminate the fault

Solar MPPT Converter PCB - Altium, LTspice, PLECS, MSP430 Sept 2025 – Apr 2026

- Diagnosed a bring-up failure on a buck-boost topology and redesigned to a synchronous buck with half-bridge driver, reducing cost and assembly risk
- Designed a 4-layer PCB for staged bring-up with scope test points, ground references, and 0-ohm bridges to isolate each subsection before full power-on
- Hand-assembled 8+ breakout boards across power stage, gate drivers, I/V sensing, and battery charging; verified with DMM and oscilloscope
- Delivered I/V sensing within 3.3V ADC range across 0–42V input using INA299A3 amplifier and resistor divider for MSP430 firmware

Education

Simon Fraser University

Bachelor of Applied Science, Sustainable Energy Engineering

Expected Sept 2027

Relevant Courses: *Embedded Computer Systems, Electronic Devices and Systems, Feedback Control Systems, Power Electronics*

Technical Skills & Interests

Design Tools: Altium Designer, SolidWorks, LTspice, PLECS, MATLAB/Simulink, GitHub

Embedded/Hardware: STM32 (Cortex-M4), Arduino (ATmega328P), MSP430, I²C, UART, SPI, PWM

Programming: C, C++, Python

Lab Tools: Oscilloscope, DMM, Function Generator, Power Supply, Soldering Iron, 3D-Printing, Wire Crimp

Interests: Street Photography, Psychology Books, Running, Hot Yoga, Bodybuilding, Hiking, Volleyball